

electron cutouts. Measurements have also been carried out in a water phantom at the depth of maximum dose as a function of the air gap g between the far end of the applicator and the phantom surface. Measurements have been compared to calculations with the Oncentra Masterplan treatment planning system.

Results & conclusion. Our results indicate that the electron beam output does not vary significantly with the insertion of an electron cutout into the beam and therefore, the monitor units determined for a given electron applicator can be used for the blocked beam as well. Monitor Units can also be safely determined by using the SSD_{eff} in order to account for the change in output and depth-dose distribution.

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DOSE-VOLUME PREDICTORS OF RADIO-INDUCED COMPLICATIONS AFTER RADIOSURGERY FOR UVEAL MELANOMA

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Introduction. Uveal melanoma (UM) is a life threatening intraocular malignant tumor. Gamma Knife Stereotactic Radiosurgery (GKSRS) is a well-assessed strategy for conservative treatment of UM providing good results for survival, local control and eye preservation.

Purpose. Our aim is to develop predictive models for radio-induced effects in UM patients treated with exclusive GKSRS.

Materials and methods. Medical records and 3D dosimetry data of critical structures of 66 patients were retrospectively reviewed. Cox's proportional hazard model was used to identify clinical and dosimetric variables as independent risk factor for most clinically relevant GKSRS-related complications: radiation vasculopathy (RV), radiation papillopathy (RP) and neovascular glaucoma (NVG), visual acuity (VA) reduction >20% (VA20%) and complete loss of basal VA (VA100%). ROC curve analysis allowed predicting cutoff values of significant variables.

Results. The 2 years incidences were: RV 10%, RP 12%, NVG 14%, VA20% 59% and VA100% 27%. A clear relationship with D1% to optic nerve (ON) was found for RP and RV. Also, for RV, the prescription isodose-ON distance (PIOND) and the anterior to equator position of the tumor resulted as protective variables. The V20 of posterior segment and tumor thickness were predictive for NVG. Multivariate analyses resulted in two variables predictive model both for VA20% and VA100%, including tumor largest diameter and D1% to ON. A second predictive model, including PIOND, was found for VA100%.

Conclusion. We found clinical and dosimetric variables to predict the risk of main side effects after GKSRS for UM. These results may provide new dose constraints to critical structures potentially able to reduce toxicities.

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ULTRAVIOLET RAY WAVELENGTH IN GAFCHROMIC XR-RV3 AND XR-SP2 FILMS

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Introduction. Gafchromic films have been used for the measurement of X-ray dose in diagnostic radiology. Correlation between color density change and ultraviolet (UV) wavelength of Gafchromic XR-RV3 and XR-SP2 were studied. UV irradiation is considered as an X-ray substitute in the double exposure technique. It is one of the correction methods for the nonuniformity error of Gafchromic XR-RV3 and XR-SP2.

Purpose. It is to determine the UV suitable wavelength of a UV light emitting diode (UV LED) that is used as a substitute for X-rays in the double exposure technique for the Gafchromic XR-RV3 and XR-SP2.

Materials and methods. UV LED that UV rays at a wavelength of 353 nm and 5-nm increments from 360 to 400 nm were used. Gafchromic XR-RV3 or XR-SP2 was irradiated with UV of different wavelengths through the 30 mm diameter of irradiation hole for 2 and 4 h, at a distance of 53 mm, respectively. Gafchromic XR-RV3 or XR-SP2 was fixed onto an acrylic plate of 3 mm thickness for assistance with scan. A circular region of interest (ROI) of 0.5 inch for measurements of pixel value was set in the center of the UV irradiation region. The mean pixel values $\pm$ standard deviation (SD) was obtained.

Results. Indicated high pixel value of Gafchromic XR-RV3 and XR-SP2 were 1276.81 ± 254.30 (395 nm) and 576.46 ± 219.75 (400 nm), respectively.

Conclusions. The irradiation of Gafchromic XR-RV3 and XR-SP2 with homogeneous UV rays by the UV-LED wavelengths of 395 or 400 nm can correct the nonuniformity error of the active layer efficiently.

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